

# cal1615 Winder Accumulator - Position Control Changes

Program P31\_WAC, revisions a01 (2026-09-23) and a02 (2026-09-24). File: AJ\_cal1615\_20260924\_a02.L5K. Base: AJ\_cal1615\_20260909\_1920.L5K.

**What changed, in one paragraph.** Before these revisions the accumulator carriage had no position setpoint while running: the winder overspeeded to pay the carriage out to empty, then handed tension control back to the load cell. Revision a02 adds a **hold-at-setpoint** mode, on by default, in which the winder speed is trimmed both ways so the carriage runs at an operator-entered position (20% shipped) and pressure sets web tension only. Revision a01 adds a separate **manual position mode** for use with the winders stopped, where a PID on the tension pressure moves the carriage to an entered position and holds it there. Both live entirely in P31\_WAC. No rungs in P29\_Winder were changed; the winder is driven through the existing p00\_wac interface tags.

## 1. Revision a02 - Hold at setpoint (winder speed owns position)

### 1.1 Principle

In tension mode the winder speed is line speed times  $(1 + p00\_wac\_speed\_offset)$ . Position is the integral of the speed mismatch between the line and the winder, so a proportional trim on position error settles exactly on the setpoint with no integral and nothing to hunt:

- Carriage above setpoint: winder faster than line, carriage empties.
- Carriage below setpoint: winder slower than line, carriage fills.
- Carriage at setpoint: winder at line speed, trim is zero.

Pressure is never used for position. `param_prs_payout` plus the recipe shift is now the running tension pressure.

`p00_wac_enabled` stays latched for the whole run, so the winder tension PID stays in software manual tracking the trim and the load-cell loop does not take over mid-roll. Torque mode is unchanged: the winder runs against a torque limit there and has no speed to trim.

### 1.2 Routine R11\_Modes - 1 new rung

| Rung | Change              | Logic                                                                                                                                                                                                                                                        |
|------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>NEW</b> Mode bit | Hold mode is active when the parameter is on, the accumulator selector is on, no A08 transducer fault, and the recipe is in tension mode.<br><br><code>XIC(param_acc_hold_mode) XIC(pb_acc_on) XIO(A08) XIC(p29r09_tension_mode) OTE(acc_hold_active)</code> |

### 1.3 Routine R22\_Fill\_Pressure - 4 rungs modified, 4 rungs added

| Rung | Change                | Logic                                                                                                                                                                                                                                                                                                                                                         |
|------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | <b>MODIFIED</b> Latch | A third branch latches <code>p00_wac_enabled</code> whenever hold mode is active, so the winder is speed-trimmed from the moment it starts, not only after a fill.<br><br><code>XIC(pb_acc_on) XIO(A08) [XIC(request_fill) , CMP(p00_wac_position_pct &gt; (param_acc_empty_pct + param_acc_empty_hyst)) , XIC(acc_hold_active) ] OTL(p00_wac_enabled)</code> |

| Rung | Change                            | Logic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | <b>MODIFIED</b><br>Unlatch        | The empty-position unlatch is blocked while holding, so the winder never hands back to the load cell mid-roll. Selector off and A08 still unlatch.<br><br><pre>[XIO(pb_acc_on) , XIO(acc_hold_active) LEQ(p00_wac_position_pct,param_acc_empty_pct) , XIC(A08) ] OTU(p00_wac_enabled)</pre>                                                                                                                                                                                          |
| 4    | <b>MODIFIED</b><br>Settled detect | The settle timer that releases the winder diameter calculation now also runs while holding, once the trim magnitude is inside the settled band. The diameter calc back-solves from line speed and winder rpm, so it must keep running as the roll builds or the trim would run off its clamp.<br><br><pre>[XIO(p00_wac_enabled) , XIC(acc_hold_active) LEQ(acc_hold_trim_abs,param_acc_hold_settle_band) ] MOV(param_acc_settle_ms,acc_settle_tmr.PRE) TON(acc_settle_tmr,?,?)</pre> |
| 20   | <b>NEW</b> Setpoint clamp         | Operator entry clamped between <code>param_man_pos_sp_min</code> (5%) and <code>param_acc_full_clear_pct</code> (70%), so the setpoint can never sit inside the full line-stop region.<br><br><pre>MOV(param_acc_hold_pos_sp,acc_hold_sp) [GRT(acc_hold_sp,param_acc_full_clear_pct) MOV(param_acc_full_clear_pct,acc_hold_sp) , LES(acc_hold_sp,param_man_pos_sp_min) MOV(param_man_pos_sp_min,acc_hold_sp) ]</pre>                                                                 |
| 21   | <b>NEW</b> Error                  | Position error, positive when the carriage is above the setpoint, and the negative trim clamp.<br><br><pre>CPT(acc_hold_err,p00_wac_position_pct-acc_hold_sp) CPT(acc_hold_trim_min,-1.0*param_acc_hold_trim_max)</pre>                                                                                                                                                                                                                                                              |
| 22   | <b>NEW</b> Speed trim             | The last word on <code>p00_wac_speed_offset</code> while holding. Error times gain, clamped both ways. This replaces the one-way payout taper written on rung 19.<br><br><pre>XIC(acc_hold_active) CPT(p00_wac_speed_offset,acc_hold_err*param_acc_hold_gain) [GRT(p00_wac_speed_offset,param_acc_hold_trim_max) MOV(param_acc_hold_trim_max,p00_wac_speed_offset) , LES(p00_wac_speed_offset,acc_hold_trim_min) MOV(acc_hold_trim_min,p00_wac_speed_offset) ]</pre>                 |
| 23   | <b>NEW</b> Trim magnitude         | Feeds the settled detect on rung 4.<br><br><pre>CPT(acc_hold_trim_abs,ABS(p00_wac_speed_offset))</pre>                                                                                                                                                                                                                                                                                                                                                                               |
| 25   | <b>MODIFIED</b><br>Payout active  | Payout overspeed is disabled while holding, so the P29 payout override cannot fight the trim.<br><br><pre>XIC(pb_acc_on) XIO(A08) XIC(winder_running) GRT(acc_taper,0.0) XIO(acc_hold_active) OTE(p00_wac_payout_active)</pre>                                                                                                                                                                                                                                                       |

#### 1.4 How the trim reaches the winder (existing P29 rungs, not changed)

| Routine       | Existing rung                                                                 | Role                                                                    |
|---------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| P29 R10 / R11 | <pre>XIC(p00_wac_enabled) MOV(p00_wac_speed_offset,p29r10_speed_offset)</pre> | Copies the accumulator trim into the winder speed offset while enabled. |

| Routine       | Existing rung                                                                                        | Role                                                                                 |
|---------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| P29 R10 / R11 | XIC(p00_wac_enabled) CPT(p29r10_pid_psn.S0, ...) ...<br>OTE(p29r10_pid_psn.SWM)                      | Keeps the load-cell PID in software manual, tracking the trim for a bumpless return. |
| P29 R15 / R16 | XIC(p29r15_stop_dwell.DN) XIO(p29r09_torque_mode)<br>CPT(p29r15_speed_factor, 1+p29r10_speed_offset) | Applies the offset to the winder speed factor in tension mode.                       |
| P29 R15 / R16 | XIC(p00_wac_payout_active) CPT(p29r15_speed_factor, 1.0+...)                                         | Payout override, now held off by R22 rung 25 while holding.                          |

### 1.5 New tags in P31\_WAC (a02)

| Tag                        | Type                   | Default | Purpose                                                                                                                   |
|----------------------------|------------------------|---------|---------------------------------------------------------------------------------------------------------------------------|
| param_acc_hold_mode        | BOOL                   | 1       | 1 = hold at setpoint. 0 = previous behaviour, pay out to empty and hand over to the load cell. Can be changed at runtime. |
| param_acc_hold_pos_sp      | REAL                   | 20 %    | Run position, percent of stroke.                                                                                          |
| param_acc_hold_gain        | REAL                   | 0.02    | Speed trim per percent of error. 0.02 gives 20% overspeed at 10% error.                                                   |
| param_acc_hold_trim_max    | REAL                   | 0.25    | Trim clamp both ways. Winder stays between 75% and 125% of line speed.                                                    |
| param_acc_hold_settle_band | REAL                   | 0.02    | Trim magnitude below this lets the diameter calc run.                                                                     |
| acc_hold_active            | alias<br>hmi_dint[9].4 |         | Hold mode in effect, for the HMI.                                                                                         |
| acc_hold_sp                | REAL                   |         | Setpoint in use after clamping.                                                                                           |
| acc_hold_err               | REAL                   |         | Position error, percent of stroke.                                                                                        |
| acc_hold_trim_min          | REAL                   |         | Negative of the trim clamp.                                                                                               |
| acc_hold_trim_abs          | REAL                   |         | Trim magnitude for the settled detect.                                                                                    |

### 1.6 Operating notes

- Doff fills start from the setpoint instead of empty, so the full line stop arrives about a fifth of a stroke sooner at a 20% setpoint.
- The loop time constant is about 13 s at 23 fpm and shortens with line speed. If the carriage oscillates at high speed, halve the gain. If it wanders, raise it.
- Set `param_acc_hold_mode` to 0 to get the previous pay-out-to-empty behaviour without a download.

## 2. Revision a01 - Manual position mode (pressure PID, winders stopped)

### 2.1 Principle

The operator enters a carriage position and requests manual position mode. A built-in PID instruction drives the tension pressure so the carriage moves there and holds. The tare pressure, at which the carriage neither rises nor

falls, is the PID bias, so the loop only supplies the difference that actually moves the carriage. The mode is permitted only with both winders stopped, because once a winder takes web the carriage position is set by winder speed and not by pressure.

## 2.2 Routine R00\_Main - 1 new rung

| Rung | Change             | Logic                                                                                                                           |
|------|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 6    | <b>NEW</b><br>Call | Runs the new routine ahead of R22 so its pressure command is ready before the sys_Ramp rung.<br><br>JSR(R23_Position_Control,0) |

## 2.3 Routine R22\_Fill\_Pressure - 1 new rung

| Rung | Change                 | Logic                                                                                                                                                                                                |
|------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12   | <b>NEW</b><br>Override | Manual mode has the last word on selected_pressure ahead of the sys_Ramp rung, so the existing accel/decel ramp still smooths it.<br><br>XIC(man_pos_active) MOV(man_pos_pressure,selected_pressure) |

## 2.4 Routine R23\_Position\_Control - new routine, 16 rungs

| Rung | Section    | Logic                                                                                                                                                                                                                                                                                                          |
|------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | Header     | NOP()                                                                                                                                                                                                                                                                                                          |
| 1    | Permissive | A running winder drops the request. The operator re-requests when needed.<br><br>XIC(winder_running) OTU(man_pos_request)                                                                                                                                                                                      |
| 2    | Permissive | Active with the HMI request, selector on, no A08, both winders stopped.<br><br>XIC(man_pos_request) XIC(pb_acc_on) XIO(A08) XIO(winder_running) OTE(man_pos_active)                                                                                                                                            |
| 3    | Setpoint   | Clamped between 5% and 90% so the carriage never drives onto a limit switch.<br><br>MOV(param_man_pos_sp,man_pos_sp) [GRT(man_pos_sp,param_man_pos_sp_max) MOV(param_man_pos_sp_max,man_pos_sp) , LES(man_pos_sp,param_man_pos_sp_min) MOV(param_man_pos_sp_min,man_pos_sp) ]                                  |
| 4    | PID setup  | Setpoint, process variable (carriage position) and bias (tare pressure).<br><br>MOV(man_pos_sp,man_pos_pid.SP) MOV(ai_lpt01_pct,man_pos_pid.pv) MOV(param_tare_pressure,man_pos_pid.BIAS)                                                                                                                      |
| 5    | PID setup  | Gains and deadband from the parameters every scan, tunable live.<br><br>MOV(param_man_pos_kp,man_pos_pid.KP) MOV(param_man_pos_ki,man_pos_pid.KI) MOV(param_man_pos_kd,man_pos_pid.KD) MOV(param_man_pos_db,man_pos_pid.DB)                                                                                    |
| 6    | PID setup  | Output limits in psi (MAXO/MINO, which also stop integral windup) and scaling so the CV tag reads in psi.<br><br>MOV(param_man_pos_prs_max,man_pos_pid.MAXO) MOV(param_man_pos_prs_min,man_pos_pid.MINO) MOV(100,man_pos_pid.MAXCV) MOV(0,man_pos_pid.MINCV) MOV(100,man_pos_pid.MAXS) MOV(0,man_pos_pid.MINS) |

| Rung | Section   | Logic                                                                                                                                                                                                                            |
|------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | PID setup | Keep the bias as entered; deadband takes effect as soon as the error is inside it.<br>OTE(man_pos_pid.NOBC) OTE(man_pos_pid.NOZC)                                                                                                |
| 8    | Bumpless  | Software manual while the mode is off.<br>XIO(man_pos_active) OTE(man_pos_pid.SWM)                                                                                                                                               |
| 9    | Bumpless  | Track the live tension pressure so the mode starts from where the carriage already is.<br>XIC(man_pos_pid.SWM) MOV(ao_prs02,man_pos_pid.S0)                                                                                      |
| 10   | Update    | P31 is in the continuous task, so the PID is paced by its own timer with .UPD held equal to the interval.<br>MOV(param_man_pos_upd_ms,man_pos_timer.PRE) CPT(man_pos_pid.UPD,param_man_pos_upd_ms/1000.0) TON(man_pos_timer,?,?) |
| 11   | Update    | XIC(man_pos_timer.DN) PID(man_pos_pid,man_pos_pid_pv,0,man_pos_pid_cv,0,0,0) RES(man_pos_timer)                                                                                                                                  |
| 12   | Output    | MOV(man_pos_pid_cv,man_pos_pressure)                                                                                                                                                                                             |
| 13   | Output    | CPT(man_pos_err,ABS(man_pos_sp-ai_lpt01_pct))                                                                                                                                                                                    |
| 14   | Output    | At-position status for the HMI.<br>XIC(man_pos_active) LEQ(man_pos_err,param_man_pos_db) OTE(man_pos_at_position)                                                                                                                |
| 15   | End       | RET()                                                                                                                                                                                                                            |

## 2.5 New tags in P31\_WAC (a01)

| Tag                         | Type                    | Default  | Purpose                                                            |
|-----------------------------|-------------------------|----------|--------------------------------------------------------------------|
| man_pos_request             | alias<br>hmi_dint[9].9  |          | HMI request, maintained bit.                                       |
| man_pos_active              | alias<br>hmi_dint[9].11 |          | Mode in effect.                                                    |
| man_pos_at_position         | alias<br>hmi_dint[9].14 |          | Error inside the deadband.                                         |
| param_man_pos_sp            | REAL                    | 20 %     | Target carriage position.                                          |
| param_man_pos_sp_min / _max | REAL                    | 5 / 90 % | Setpoint clamp.                                                    |
| param_man_pos_kp            | REAL                    | 1.0      | Controller gain, psi per percent of stroke (dependent gains form). |
| param_man_pos_ki            | REAL                    | 0.25     | Reset time, minutes per repeat. 0 = no integral.                   |
| param_man_pos_kd            | REAL                    | 0        | Rate time, minutes.                                                |

| Tag                                                                                                         | Type                | Default        | Purpose                  |
|-------------------------------------------------------------------------------------------------------------|---------------------|----------------|--------------------------|
| param_man_pos_db                                                                                            | REAL                | 0.5 %          | Hold deadband.           |
| param_man_pos_prs_min / _max                                                                                | REAL                | 18 / 80<br>psi | Pressure command limits. |
| param_man_pos_upd_ms                                                                                        | DINT                | 100            | PID update interval.     |
| man_pos_pid, man_pos_pid_pv, man_pos_pid_cv,<br>man_pos_pressure, man_pos_sp, man_pos_err,<br>man_pos_timer | PID, REAL,<br>TIMER |                | Working tags.            |

### 3. Verification performed on each revision

---

Each revision was built from the previous file by exact-string replacement with every anchor asserted to match once, then checked: CRLF line endings with zero bare LF, every rung comment followed by exactly one rung, all ROUTINE/PROGRAM/TAG/DATATYPE/MODULE/AOI blocks paired, every identifier in the touched routines resolving to a declared tag, every tag datatype existing, and a diff against the previous revision showing only the intended lines. a01: 124 lines added, none removed. a02: 58 lines added and 4 removed, the 4 removed lines being the modified rungs replaced in place.